FormForge 2.0

An Information-Aware Adaptive Questionnaire Platform with Privacy-Preserving AI



Ask fewer, better questions

[STUDENT NAME] | [REGISTER NUMBER]

Guide: Prof. / Dr. [Guide Name]

Master of Computer Applications | SCORE, VIT Vellore

Agenda

01

Context

Review 1 feedback, problem, objectives

02

Literature

2024-2026 research and the gap

03

Design

Architecture, UML, data model, algorithms

04

Build

Implementation, testing and live demo

05

Results

Static vs rule-based vs adaptive, cold start, ablation

06

Close

Cost, conclusion and future scope

From Review 1 to FormForge 2.0

Review 1: FormForge 1.0

- Multi-tenant form builder with RBAC workspaces
- Drag-and-drop builder, public form links
- Webhooks, e-mail workflows, SSE live counter
- Feedback: "nothing new, built many times before"

Now: research-driven 2.0

- Adaptive engine picks the next question per respondent
- A free LLM generates forms with metadata
- Automatic questionnaire quality analysis
- Synthetic cold start, PII masking, insights
- Measured against static and rule-based forms

The Problem with Static Forms



Same list for everyone

A student who never met a mentor still answers five mentoring questions



Long forms get abandoned

In our simulation about 30% of respondents left the 16-question static form midway



Badly written questions

46% of 250 items in locally designed surveys were not recommended for use (Reed et al., 2024)



AI usually means cloud

Paid APIs fail on quota and send personal data off the machine

Research question

Can a form decide, for each respondent, which question is worth asking next, so that useful information is collected with fewer questions and less effort?

...while running at zero cost and keeping response data on the machine.

Objectives

AI form generation

Plain-language prompt to a structured form with metadata, via a free LLM

Quality analyzer

Flag double-barrelled, leading, ambiguous and redundant questions; estimate time

Adaptive engine

Score every unanswered question and ask the one with the highest utility

Cold start

Warm start new forms with synthetic respondents whose weight decays

Privacy and insights

Mask PII before any model call; summarise answers into topics and sentiment

Evaluation

Compare with static and rule-based forms on seven metrics

Literature Survey: Key Recent Work

Work	Focus	Key finding
Bachmann et al., 2024	Adaptive questionnaires	74% accuracy vs <40% for fixed short forms
Bachmann et al., 2025	GPT-4 synthetic cold start	RMSE 0.420 to 0.297; gain fades after ~175 users
Hu et al., 2024	Information-gain questioning	+38.1% task success with fewer questions
Li et al., 2025	Adaptive testing (ATLAS)	Up to 90% fewer items at same precision
Adhikari et al., 2025	LLM questionnaire generation	Generated items clearer and more specific
Reed et al., 2024	Quality of local surveys	46% of 250 items not recommended
Lukauskas & Šarkauskaitė, 2026	Audit of synthetic respondents	+0.84 SD acquiescence; poor transfer
Nayyeri & Rasoolzadegan, 2025	Local LLM for web forms	Local model avoids cloud data leakage

15 references in the report, 2024-2026, all with verified DOI or arXiv links

Research Gap and Our Answer

Gap in existing work	FormForge 2.0
Adaptive questionnaires exist only for fixed instruments	**Adaptive selection inside a general-purpose form builder**
Generators produce wording, not machine-readable structure	**Every generated question carries target, objective, time**
Selection optimises information alone	**Utility adds burden, redundancy and coverage**
Cold start studied with large paid models only	**Synthetic warm start with a free model and a decaying weight**
AI features send raw responses to the cloud	**PII masking before any model call; local model one setting away**

Solution Overview

Six intelligence modules added on top of the existing FormForge platform



AI Generator

Prompt to draft form

Creation



Quality Analyzer

Score + issues

Creation



Adaptive Engine

Next best question

Answering



Cold-start Simulator

Synthetic warm start

Answering



PII Redactor

Mask before AI

Analysis



Response Intelligence

Topics, sentiment

Analysis

Foundation (FormForge 1.0)

Workspaces + RBAC | JWT auth | drag-and-drop builder | public forms | webhooks with HMAC + retries | SSE live counter | analytics | CSV export

Tools and Technologies

Frontend

dnd-kit, Zustand, TanStack Query, Recharts


Next.js 14


React 18


Tailwind

Backend

zod validation, JWT, bcrypt, node-cron


Node.js


Express


TypeScript

Data

JSONB fields, migrations, Compose


PostgreSQL 16


Prisma 6


Docker

AI

Free tier, OpenAI-compatible, swappable


Gemini API


flash-lite


Ollama (opt.)

Research

Simulation, EM, evaluation


Python


NumPy


Matplotlib

Tooling

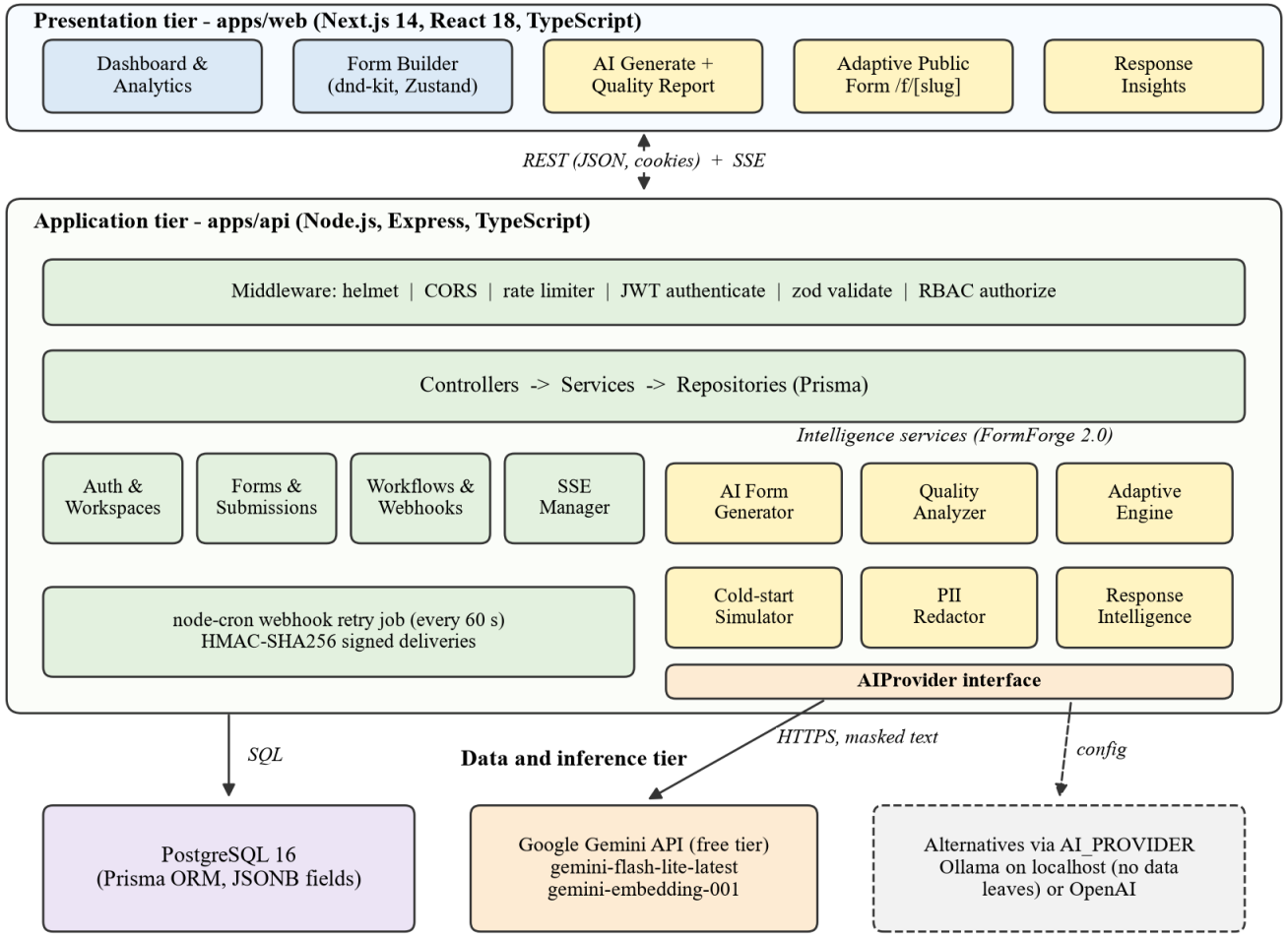
Monorepo, strict TS, 124 tests, CI


Turborepo


Vitest


zod

System Architecture



Three tiers

Next.js client, Express API, PostgreSQL + Gemini API

Fixed middleware chain

helmet, CORS, rate limit, JWT, zod, RBAC

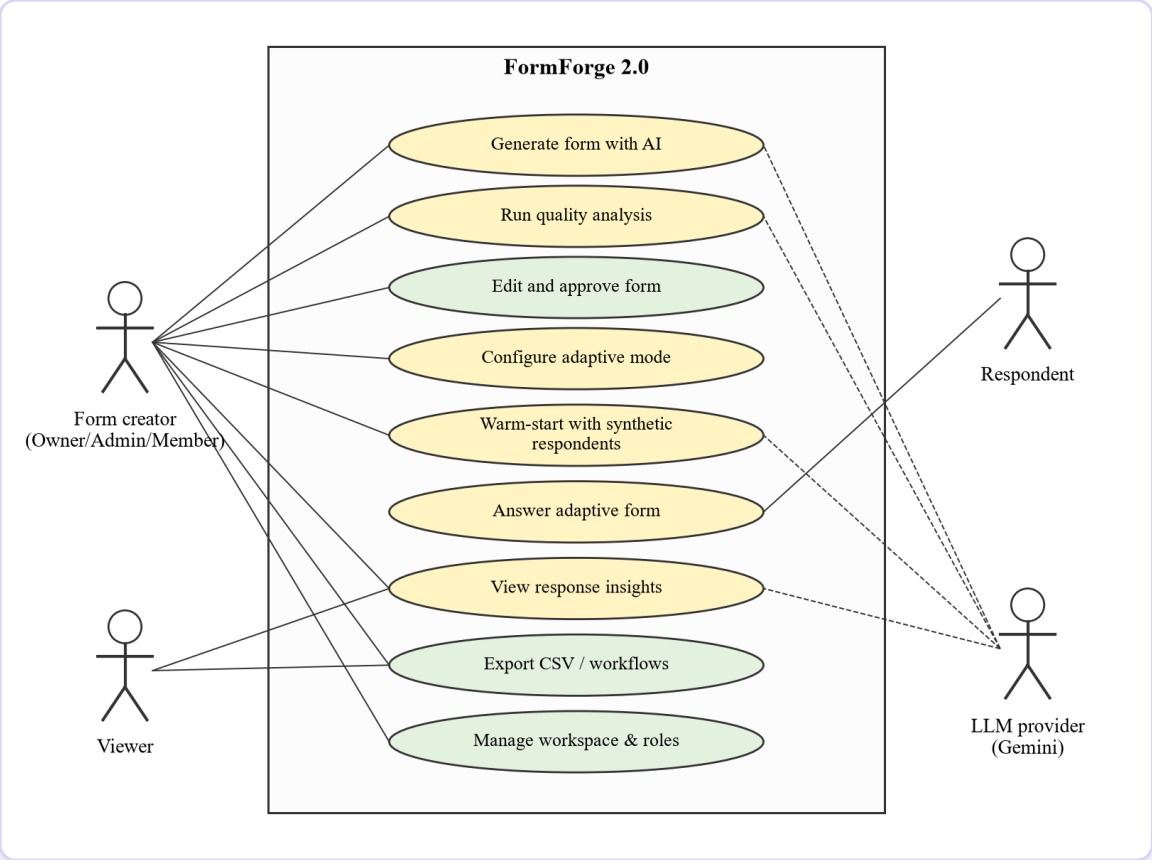
Layered API

routes, controllers, services, repositories

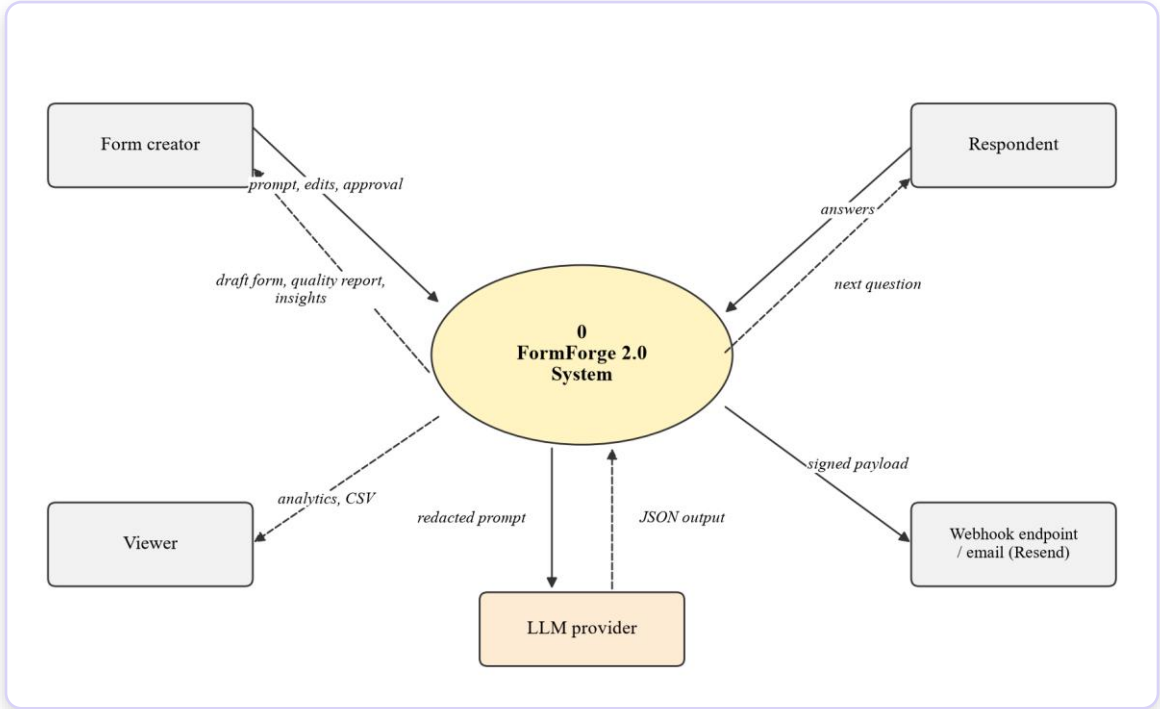
One AIProvider interface

Gemini free tier by default, masked text only

Use Case and Data Flow Diagrams

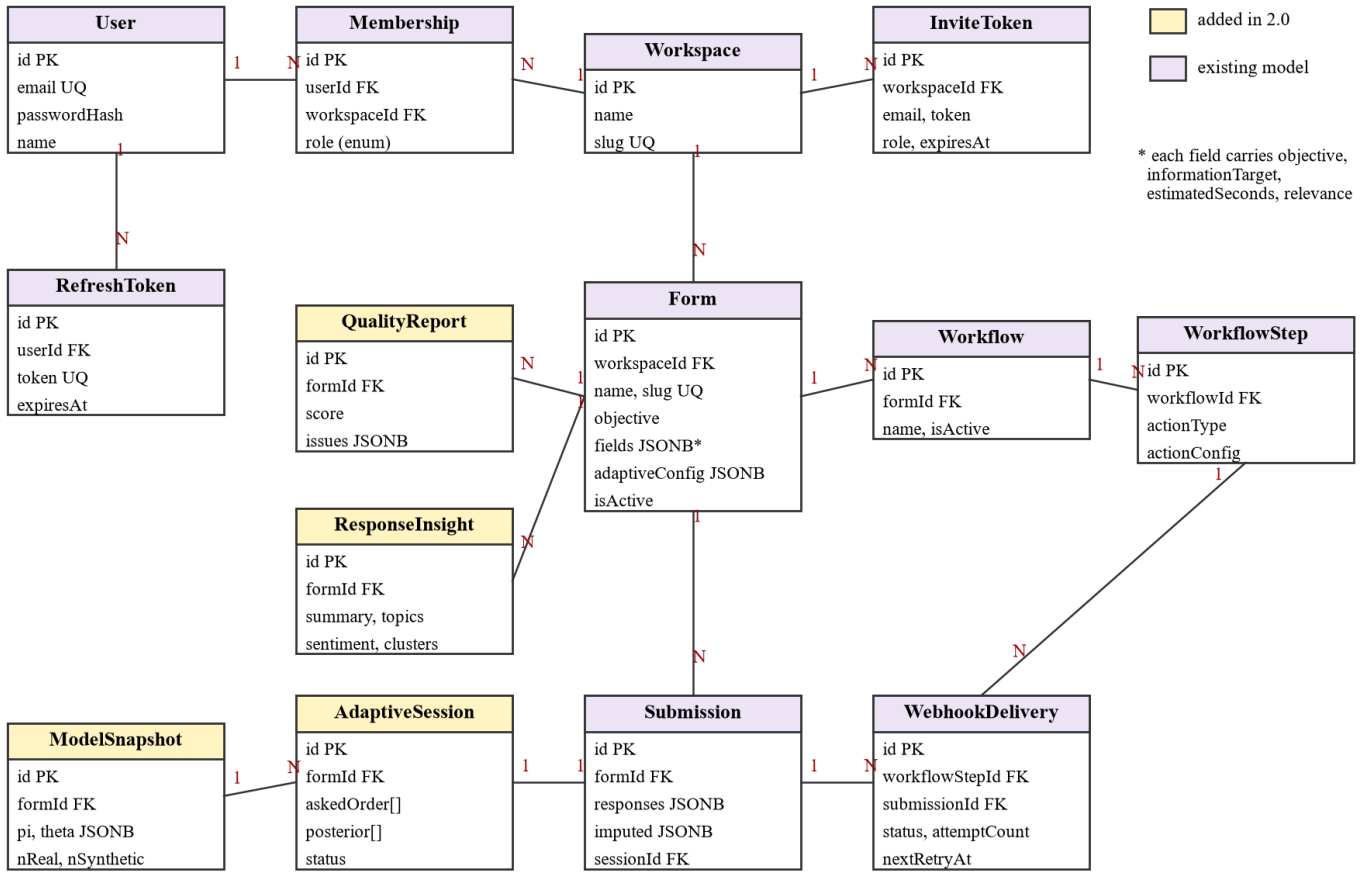


Use case: creator, viewer, respondent; LLM provider as secondary actor



DFD level 0: only redacted text flows to the model; only JSON returns

Data Model



9 existing models kept unchanged

Only three columns added: Form.objective, Form.adaptiveConfig, Submission.imputed

QualityReport

score, issues, time

AdaptiveSession

asked order, posterior

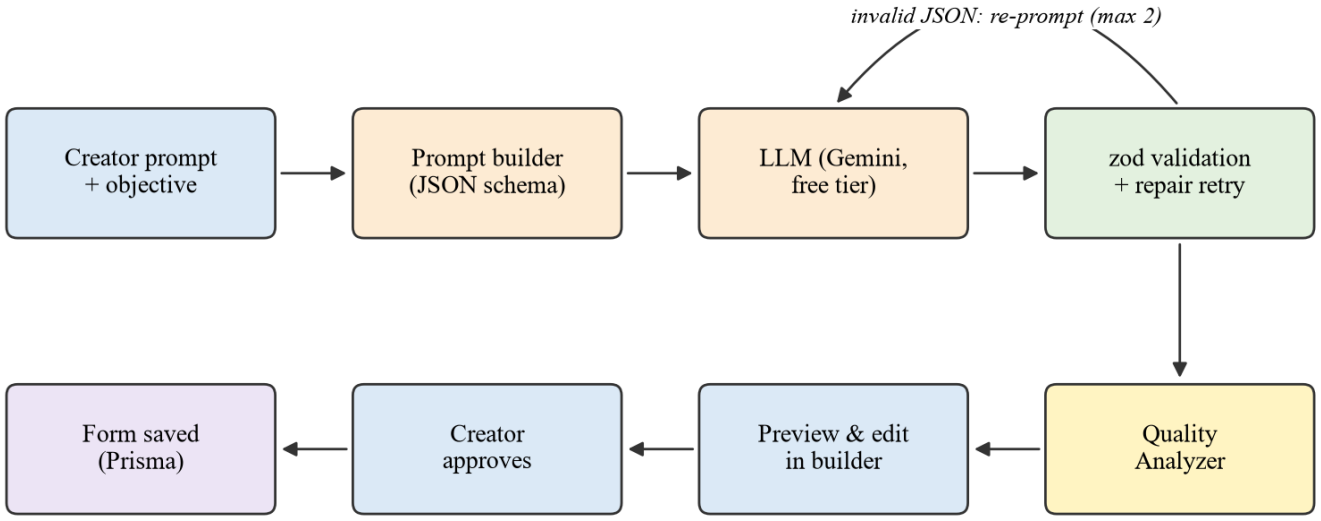
ModelSnapshot

pi, theta, nReal, nSynthetic

ResponseInsight

summary, topics, sentiment

AI Form Generation with a Free LLM



The AI produces a proposal only; nothing is written to the database without human approval

Human in the loop: the model returns a proposal only. It opens in the existing builder and nothing is saved until the creator approves.

Every question carries metadata

```
{
 "label": "Mentor feedback?",
 "type": "RATING",
 "required": false,
 "objective": "Mentoring",
 "informationTarget":
     "mentor_quality",
 "estimatedSeconds": 12,
 "relevance": 0.9
}
```

Used later by the analyzer and the adaptive engine

Form Quality Analyzer

Double-barrelled

"How satisfied were you with your mentor and project?"

Leading

"How much did you enjoy our excellent service?"

Ambiguous

"Do you exercise regularly?"

Redundant

"How happy..." vs "How satisfied..."

Readability

Flesch below 50 (6+ words) or over 25 words

Burden

Required ratio over 70%; time over 5 min

Quality score

Score = max(0, 100 - Σ w(issue))

- 12 for double-barrelled or leading
- 8 for ambiguous or redundant
- 5 for readability
- 6 for required ratio or time

Rules run instantly in the builder; an optional AI review adds depth

Completion time = base time by field type + reading time (3.5 words/s) + 0.6 s per option

Adaptive Engine: Choosing the Next Question

$$U(q) = \alpha \cdot IG(q) + \beta \cdot Rel(q) + \gamma \cdot Cov(q) - \delta \cdot Bur(q) - \varepsilon \cdot Red(q)$$

IG

Information gain

Expected drop in uncertainty over all unanswered questions

$\alpha = 0.40$

Rel

Relevance

How central the question is to the form objective

$\beta = 0.15$

Cov

Coverage

1 if its information target is still unknown

$\gamma = 0.20$

Bur

Burden

Estimated answering time, normalised

$\delta = 0.10$

Red

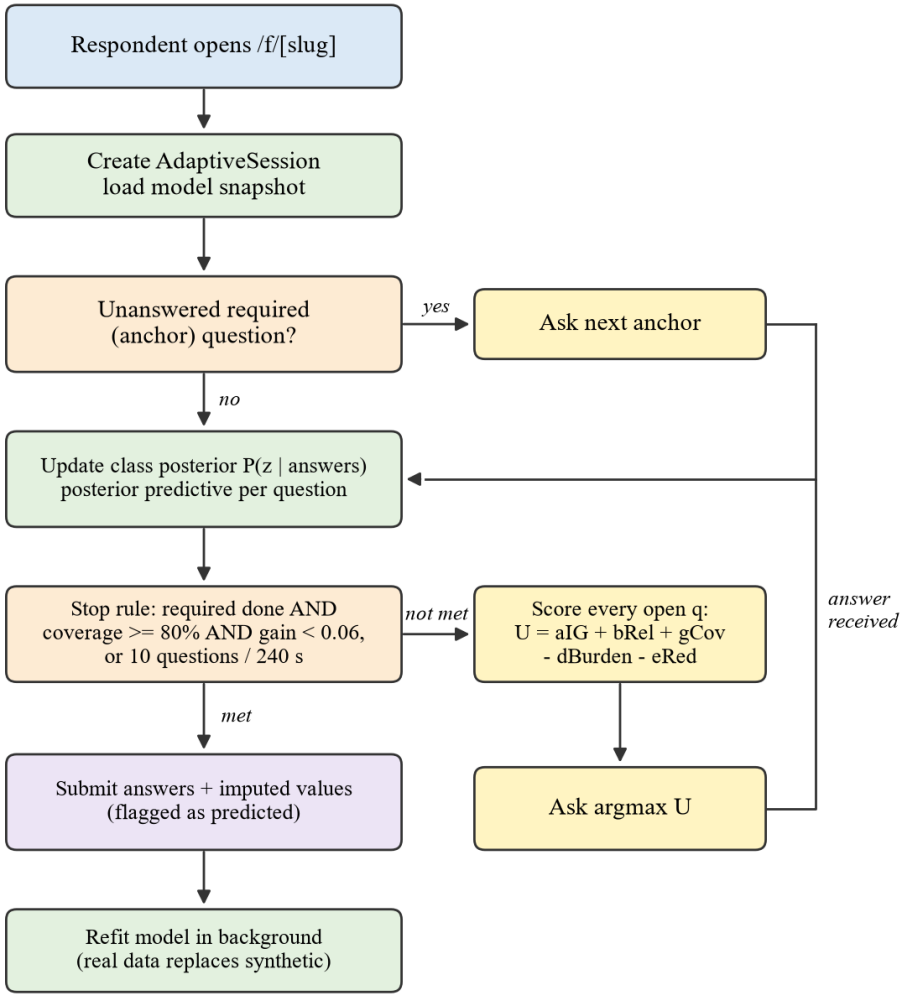
Redundancy

Max normalised mutual information with answered questions

$\varepsilon = 0.30$

Respondent model: latent class model (K = 6) fitted with EM. Posterior over classes after each answer gives predictions for every unasked question.

Adaptive Engine: Flow and Safety Rules



Anchors first

Required questions are asked first, in the creator's order

Never repeat

No question asked twice; never more than 10 or 240 s

Stop rule

Coverage ≥ 80% and remaining gain < 10% of maximum

Honest output

Skipped answers predicted, stored and labelled separately

Hidden scores

Respondents never see utilities or predictions

Cold Start with Synthetic Respondents



New form

0 real responses

Personas

The LLM writes varied respondent profiles

Synthetic answers

Options shuffled per persona, letter-labelled

Weighted fit

EM on synthetic + real data

Real data takes over

Synthetic weight decays

Synthetic weight

$$w(n) = 25 / (25 + n)$$

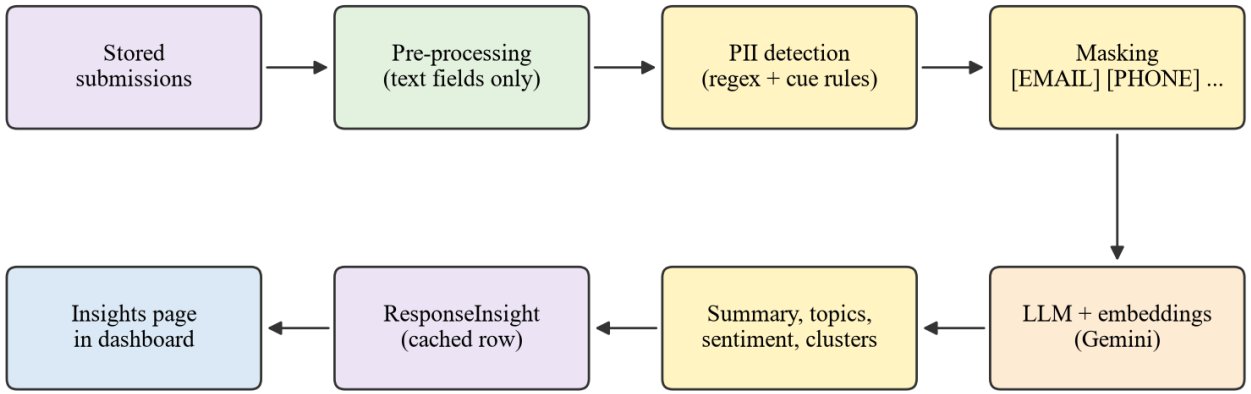
n = real responses. Full weight at 0, half at 25, one fifth at 100.

Used only to initialise, never as data

LLM respondents show low variance and an acquiescence shift of +0.84 SD (Lukauskas & Šarkauskaitė, 2026; Bisbee et al., 2024). Synthetic rows are kept apart and never appear in responses, exports or insights.

Privacy Layer and Response Intelligence

Raw personal data never reaches a model: the provider only ever sees masked text



Recomputed in the background after 20 new submissions or on demand; the page reads the cached row

Masking example

```
IN   Reach me on 7012345678 or at a.k@vit.ac.in
OUT  Reach me on [PHONE] or at [EMAIL]
```

Insights from open answers

- Summary, topics with counts, sentiment, clusters per question
- Computed in the background every 20 responses
- Detects email, phone, URL, IP, Aadhaar, PAN, register no., names

Prompt-injection safeguards

- Answers sent as delimited data, never as instructions
- Model returns answer numbers, not quotes
- Insights are display-only; never trigger actions

Implementation and Testing

95

unit tests, incl. 3 parity tests vs Python

29 / 29

API tests on a real database

15

new API endpoints

7,500

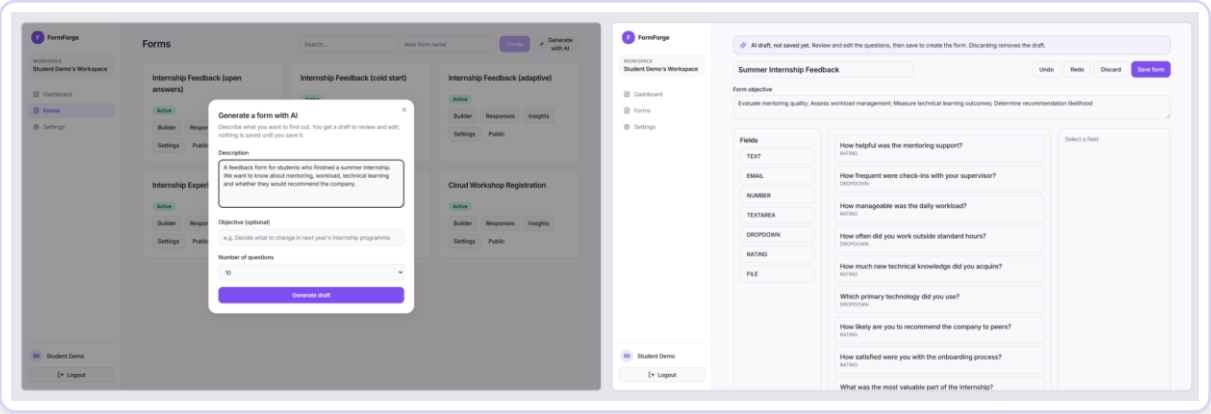
simulated sessions, 0 rule violations

Test case	Result
Duplicate e-mail registration	409 Conflict
Access another workspace	403 Forbidden
Missing required field	400, field named
6th public submit in a minute	429 Rate limited
Unpublished form by slug	404
Webhook workflow on submit	Delivery queued, HMAC signed
Adaptive session limits	No repeats, limit kept, anchors first
PII in insights	Masked; nothing personal sent

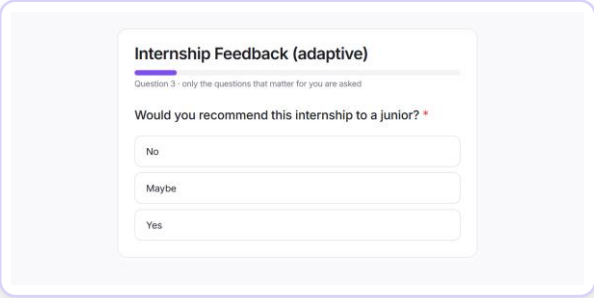
Issues found and fixed

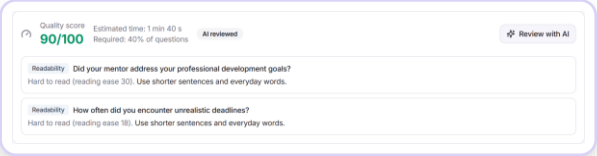
- Optional schema fields rejected: made required + nullable
- Gemini schema limits: structure-only schema
- Redundant early pick (98%): anchors first, now 0%
- CSV labels with commas: RFC 4180 writer
- Model change mid-session: snapshot pinned

Working System

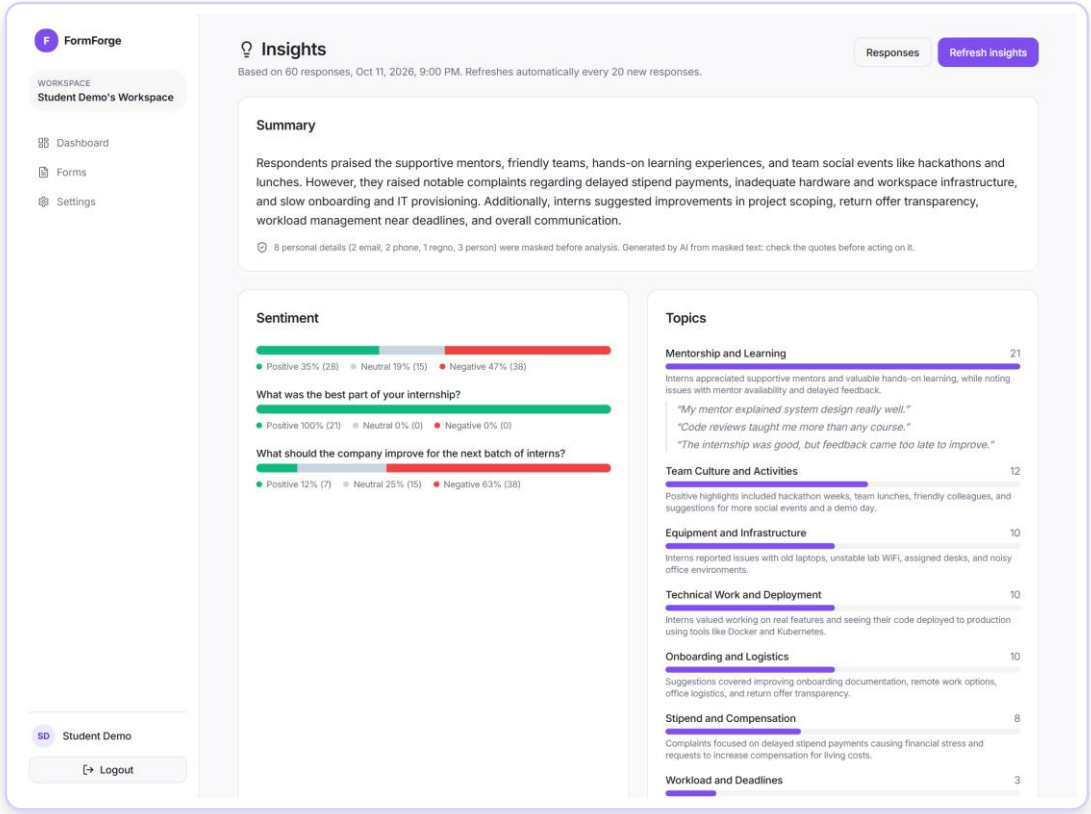


AI draft in 7 s, opened in the builder





Respondent view | quality report



Insights on masked answers: summary, sentiment, topics

Describe form → **AI draft + quality report** → **Edit and approve** → **Publish adaptive** → **Insights**

Experimental Setup

16-question form

Internship feedback: 3 required anchors, 1 deliberately redundant question

Simulated respondents

4 latent groups, correlated traits, noise; true answer known for every question

Break-off model

Quit hazard grows with elapsed time; same patience for all systems (paired)

Protocol

Model trained on 3,000; tested on 1,500; repeated with 5 seeds

Three systems compared

- Static: all 16 questions in order
- Rule-based: hand-written skip logic
- FormForge adaptive: utility engine, max 10 questions

Seven metrics

- Questions asked, completion time, completion rate
- Information coverage, redundant questions
- Prediction accuracy, targets per question

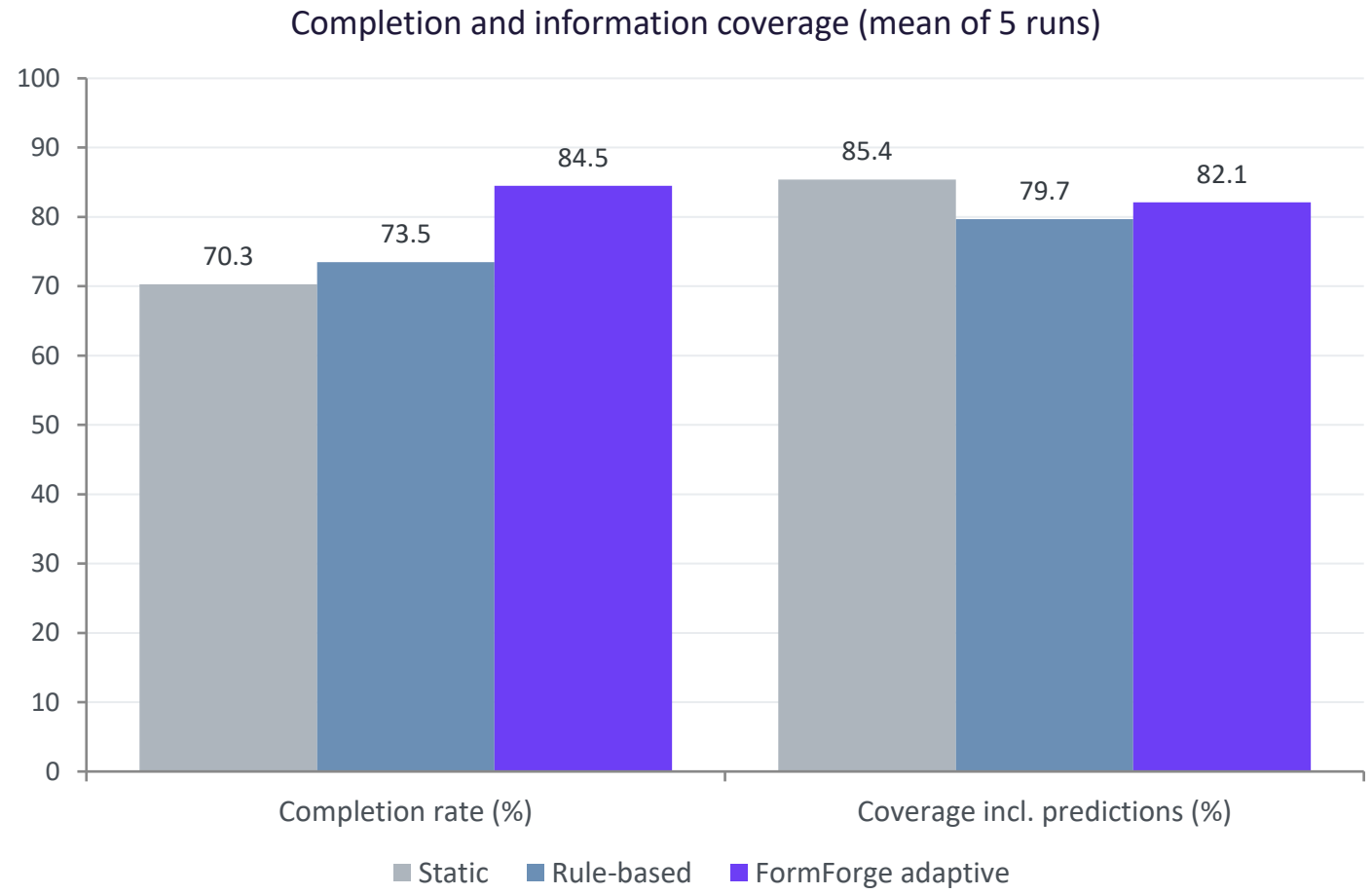
Adaptive vs Static vs Rule-Based

–35% — **questions asked**
13.7 → 8.9

–35% — **completion time**
146 s → 95 s

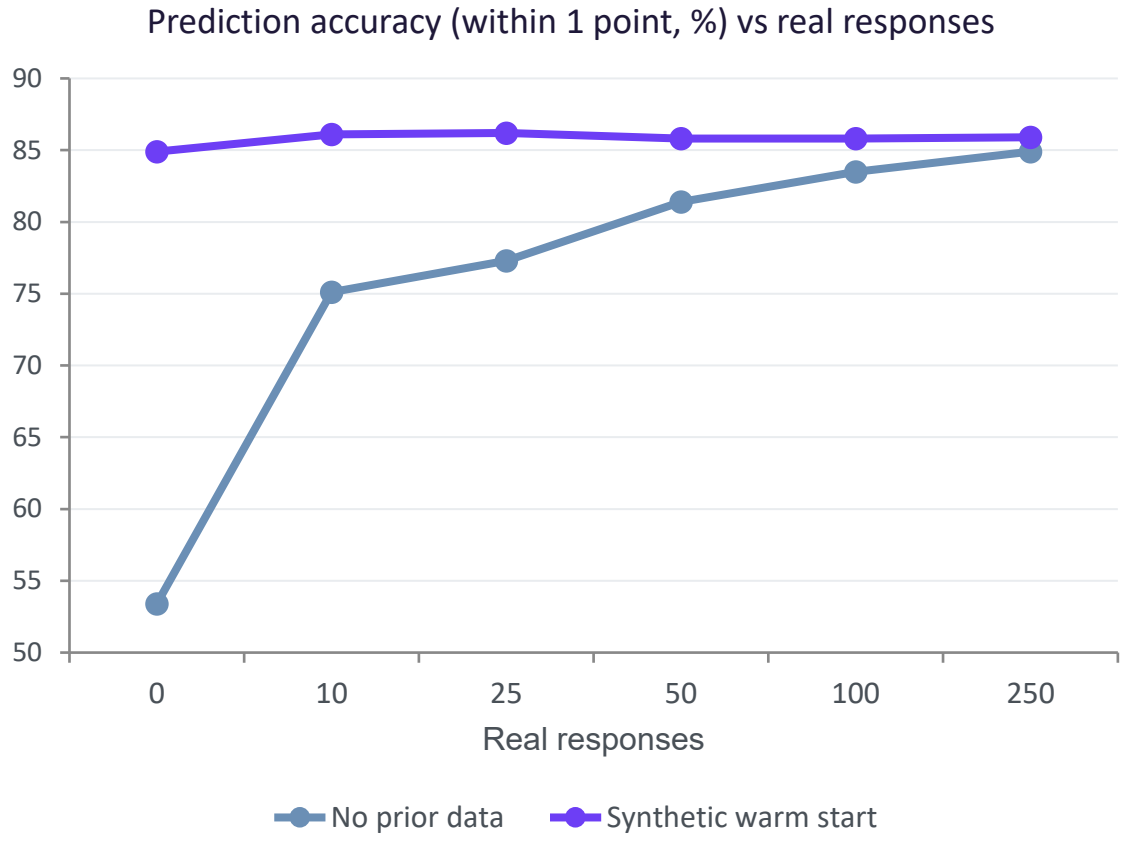
+14.1 pts — **completion rate**
70.3% → 84.5%

0 — **redundant questions**
static asked 0.31 per person

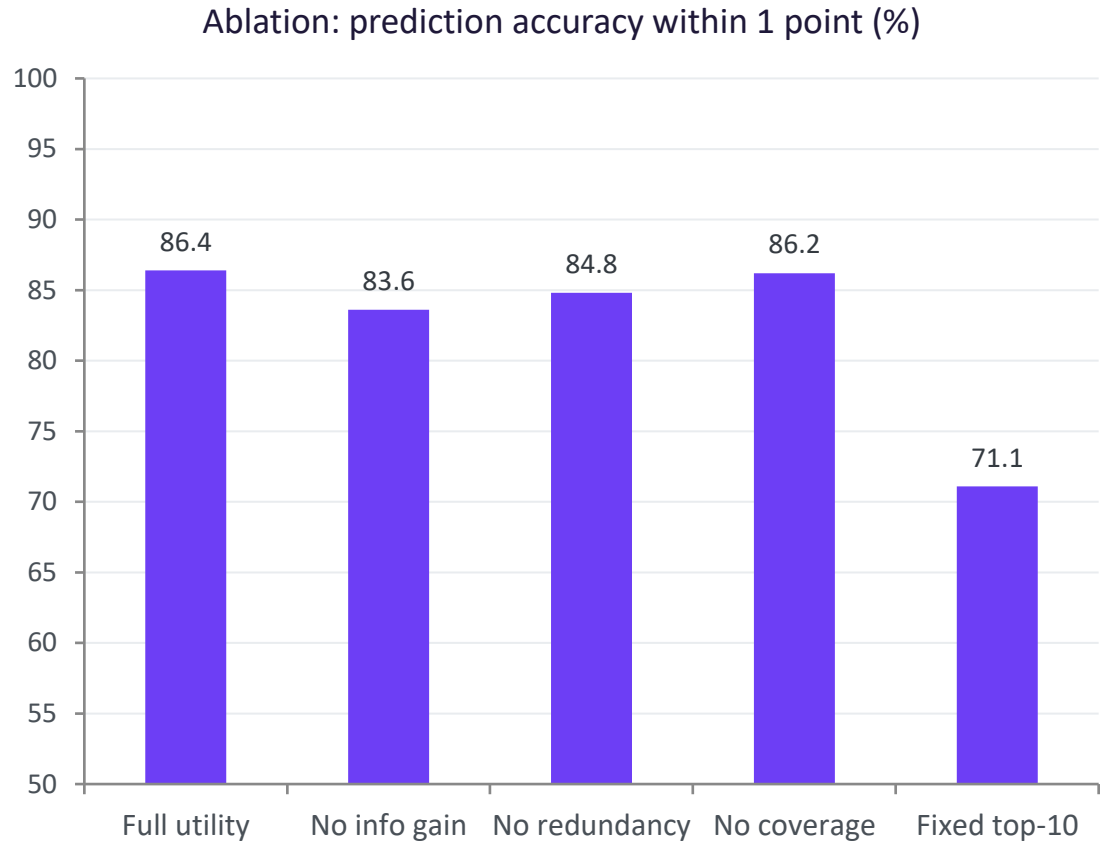


Same information, fewer questions: adaptive coverage 82.1% vs 85.4% static, at 1.53 vs 0.93 targets per question

Cold Start and Ablation

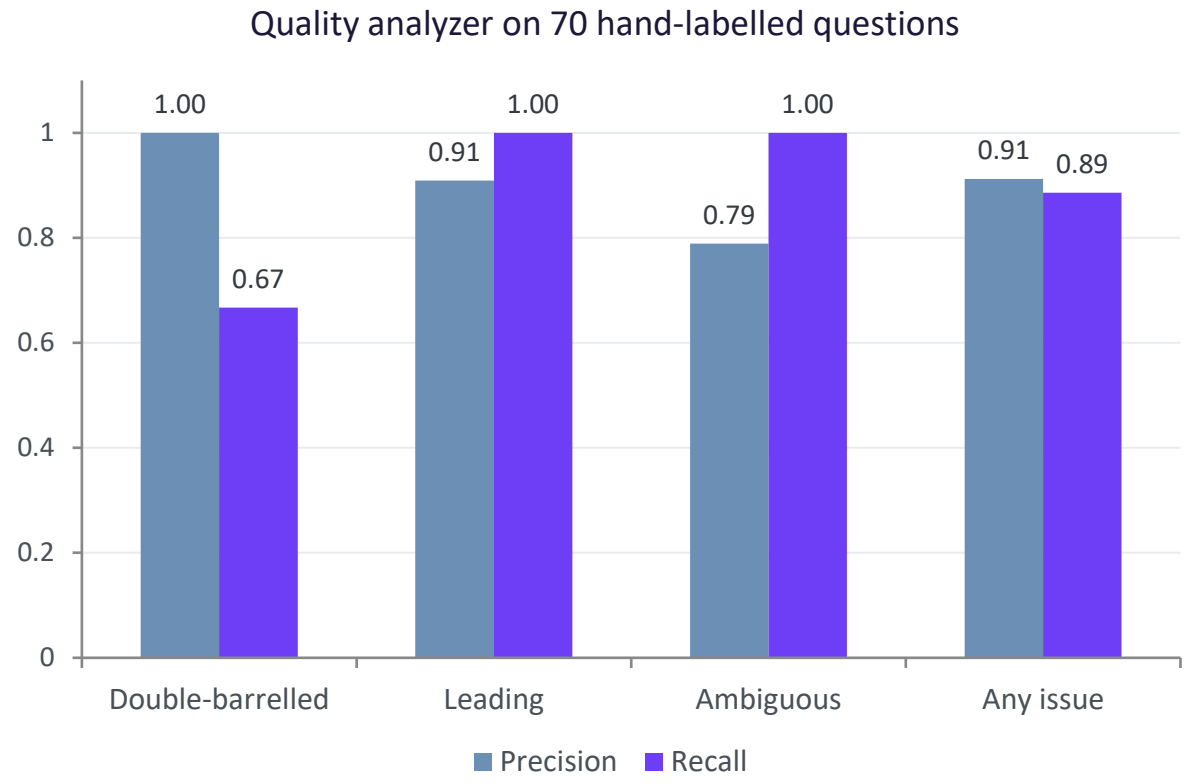


Simulation: 53% → 85% before any real response. Live check with 40 Gemini personas: 56% → 73% (real data: 82%)



The respondent model matters most: a fixed top-10 order asks 0.33 redundant questions per person vs 0 for the engine

Quality Analyzer, PII Masking and Cost



Accuracy 90% for flagged vs clean questions

F1 0.91

PII detection

20 of 23 entities masked; precision 0.95, recall 0.87

₹0

recurring cost

Free tier, no API bills

- Gemini free tier for generation and embeddings
- PostgreSQL + Node.js in Docker, all open source
- Only masked text leaves the machine; Ollama keeps all local

Conclusion and Future Scope

What we showed

- 35% fewer questions and 35% less time with almost the same information
- AI generation and quality checks at zero cost
- Synthetic warm start works for brand-new forms
- Privacy by design: masking before every model call

Future scope

- Field study with students: static vs adaptive
- Learn utility weights per form (contextual bandits)
- Grow the question bank from open answers
- Local NER model for stronger PII masking
- Adaptive selection for free-text questions

Thank you | Questions?